

Algebra/Topology Seminar

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CONJUGATOR LENGTHS AND ISOPERIMETRIC FUNCTIONS

Thursday, September 24, 2026
3:00 p.m. in Massry B008

ABSTRACT. The Dehn function of a finitely presented group is the smallest isoperimetric function of the Cayley complex associated to one of its finite presentations. This invariant gives a geometric interpretation of a fundamental algorithmic notion, providing an upper bound on the complexity of the group's Word Problem.

The conjugator length function of a finitely generated group is the function f so that $f(n)$ is the minimal upper bound on the length of a word realizing the conjugacy of two words of length at most n . Analogous to above, this invariant gives a geometric interpretation of an algorithmic notion, providing a measure for the complexity of a direct approach to the group's Conjugacy Problem.

We show that these two notions are independent in a very strong sense, exhibiting finitely presented groups whose Dehn and conjugator length functions are equivalent to just about any 'reasonable' pair of functions. In particular, we find a finitely presented group with quadratic conjugator length function and undecidable Word Problem. Moreover, we take a large step in describing all functions which can be realized as the conjugator length function of a finitely presented group.

This is joint work with Conan Gillis.